

Foundations of the Λ -algebra: the Γ_p -tower, and the relations among p -adic Apéry limits

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Abstract

Let (A, B) be a sporadic Apéry-like pair of weight w and character χ . The p -adic Apéry limit $\Lambda_a = \lim_s \chi(p)^s p^{ws} B(ap^s)/A(ap^s)$ exists in $\mathbb{Q}_p$, is not a $\mathbb{Q}$ -affine function of any single p -adic zeta value, and is assembled out of the Kazandzidis limits $c_p(A, B) = \lim_t \binom{Ap^t}{Bp^t}$; this is the content of the companion Paper C. Here we ask what these constants *are* as an algebra.

We introduce the Γ_p -tower product $\gamma_p(x) = \prod_{k \geq 1} \Gamma_p(xp^k)$ and the tower Beta $\beta_p(B, D) = \gamma_p(B + D)/(\gamma_p(B)\gamma_p(D))$, and prove $c_p(A, B) = E_p(A, B) \beta_p(B, A - B)$, where E_p is an explicit finite monomial in p and in Morita Γ_p -values at positive integers. This single computation reproves the existence of $c_p(A, B)$, isolates its transcendental part, and makes the relation problem combinatorial: writing $\delta(A, B) = [A] - [B] - [A - B]$ in the free abelian group on $\mathbb{Z}_{>0}$, we prove $\log \beta_p(B, D)$ depends on (B, D) only through δ , that δ is onto the augmentation-type subgroup $\ker(\sum c_x \mapsto \sum c_x x)$, and hence (**Rigidity Theorem**) that a multiplicative relation among the tower Betas either lies in the lattice $\ker \delta$ — generated, in the verified range, by binomial symmetry and trinomial revision — or produces a nontrivial vanishing p -adic linear form $\sum_{m \text{ odd} \geq 3} v_m \zeta_p(m) p^m / (m(1 - p^m)) = 0$ with explicit integer coefficients. Under the resulting hypothesis (IND_p) , the group generated by the tower Betas is free abelian of countably infinite rank. We report the exclusion ledger for (IND_p) : nothing below the noise floor at $p = 5, 7, 11, 13$ for $x \leq 8$ at 26 and at 38 certified p -adic digits, with the best heights scaling with the precision, as noise does.

We then define the Λ -algebra $\mathcal{L}_p$ — the closure of the algebra generated by Γ_p -values, tower products and the tower-harmonic constants $h_p(a)$ — and *prove* the block decomposition of Λ_a that Paper C established block by block only, thereby proving $\Lambda_a \in \mathcal{L}_p$ for the classical Apéry pair and every tower base a . Finally we state the archimedean/ p -adic asymmetry precisely: by the Transfer Principle of Paper D the two completions share the rational skeleton of the construction exactly, yet the archimedean Apéry limits span a one-dimensional $\mathbb{Q}$ -space per family while the p -adic ones require, under (IND_p) , infinitely many independent generators.

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1 Introduction

1.1 The object

Let $A(n)$ be one of the fifteen sporadic Apéry-like binomial sums [25, 11, 19, 23] and $B(n)$ the second solution of its recurrence, normalised by $B(0) = 0$, $B(1) = 1$. Each pair carries a weight $w \in \{2, 3\}$ and a quadratic (possibly trivial) Dirichlet character χ ; when the archimedean Apéry limit $\lim_n B(n)/A(n)$ exists it is a rational multiple of $\zeta(w)$ or of $L(\chi, w)$ [6, 10, 1]. Along a p -power tower $n_s = ap^s$ with $p \nmid a$ the normalised ratios converge p -adically, and

$$\Lambda_a := \lim_{s \rightarrow \infty} \chi(p)^s p^{ws} \frac{B(ap^s)}{A(ap^s)} \in \mathbb{Q}_p \quad (1.1)$$

is the *p -adic Apéry limit*. Its existence, its convergence rate, the proof that it is not a $\mathbb{Q}$ -affine function of $\zeta_p(3)$ or $\zeta_p(5)$ at accessible heights, and a block decomposition of it in terms of the

$$c_p(A, B) := \lim_{t \rightarrow \infty} \left(\frac{Ap^t}{Bp^t} \right) \in \mathbb{Q}_p \quad (A > B \geq 1) \quad (1.2)$$

are the content of the companion Paper C [3]. The present paper takes those results as given and asks the next question: *what algebra do these constants generate, and what are its relations?*

1.2 What this paper adds, and what it imports

This is deliberately a foundations paper for a single class of constants, and its debts should be stated before its results.

Imported, and used as a black box.

- From Paper A [1]: the twisted descent law (LB_w^χ) , $v_p(p^w B_n A_q - \chi(p) B_q A_n) \geq w$ with $q = \lfloor n/p \rfloor$, which makes (1.1) converge; and the table of the fifteen families with their weights, characters and archimedean limits.
- From Paper C [3]: the existence theorem for Λ_a , the rate law $\min(3, r(\chi, w))$, the exclusion theorem (no low-height $\mathbb{Q}$ -affine relation to $\zeta_p(3)$ or $\zeta_p(5)$), the exact closed form of the tower-harmonic constant $h_p(a)$, and the block decomposition of Λ_a , there established block by block as a Finding.
- From Paper D [4]: the Transfer Principle — the archimedean sum and the Volkenborn integral of one rational function are linear forms with *the same* rational coefficients. We use it in §5 and reprove nothing of it.

New here.

- (A) *The Γ_p -tower packaging* (§2). We introduce $\gamma_p(x) = \prod_{k \geq 1} \Gamma_p(xp^k)$ and prove in one computation (Theorem 2.5) both that $c_p(A, B)$ exists and that

$$c_p(A, B) = E_p(A, B) \cdot \beta_p(B, A - B), \quad \beta_p(B, D) = \frac{\gamma_p(B + D)}{\gamma_p(B)\gamma_p(D)},$$

with E_p elementary. This separates the arithmetic of c_p into a finite Γ_p -part and a single transcendental factor, and it is the form on which everything downstream runs. It also discharges, for these constants, an orientation remark left open in [2, §2.4].

- (B) *Rigidity* (§3). Multiplicative relations among the β_p are governed by a divisor map into a free abelian group (Theorem 3.6). Either a relation is *binomial* — an element of the lattice $\ker \delta$, which in the verified range is generated by symmetry and trinomial revision — or it certifies a nontrivial vanishing p -adic linear form in the odd Kubota–Leopoldt zeta values.
- (C) *The Λ -algebra* (§4). We define $\mathcal{L}_p$ and *prove* the block decomposition of $\tilde{A}_a = \lim_s A(ap^s)$ and of Λ_a for the classical Apéry pair and every a with $p \nmid a$ (Theorems 4.6, 4.8), upgrading Paper C’s Finding to a theorem and giving $\Lambda_a \in \mathcal{L}_p$.
- (D) *The asymmetry* (§5), stated as a precise contrast rather than a slogan: shared rational skeleton, one-dimensional archimedean constant space, infinite-rank p -adic one.
- (E) *An exclusion ledger for (IND_p)* (§6) and a programme (§7).

1.3 Two remarks on scope

The rigidity theorem is about the constants $c_p(A, B)$, which are the building blocks; it is not a determination of the relations among the Λ_a themselves, which are infinite assemblies of blocks. Paper C reports that no $\mathbb{Q}$ -linear relation among Λ_a 's, across bases or across families, is visible at accessible heights; the present paper explains why one should not expect the question to be finite-dimensional, and isolates the one hypothesis (IND_p) on which a clean answer for the blocks depends.

Second: (IND_p) is a p -adic independence statement about odd zeta values. It is out of reach — no $\zeta_p(2k+1)$ is known to be irrational for a single prime $p \geq 5$ [9, 18]. We therefore state it as a hypothesis, record what it buys, and report the numerical exclusions honestly.

1.4 Evidence classes

Every statement carries its class, printed with it.

[PROVED]

A complete proof is given here.

[IMPORTED]

Proved in a cited companion paper or in the literature, and used here without reproof. The citation is given with the statement.

[VERIFIED: range]

A finite exact computation over the stated range at the stated p -adic precision. This is evidence, never proof.

[OPEN]

Open, with its sharpest known formulation.

Statements labelled THEOREM, PROPOSITION, LEMMA, COROLLARY are [PROVED] or [IMPORTED] as marked; statements labelled FINDING rest on [VERIFIED:]evidence only and are never used as inputs to a proof. All computations reported here use exact integer or rational arithmetic; no floating point occurs at any stage, and every p -adic digit is derived from an exact rational or an exact residue. Programs are listed in Appendix A.

1.5 Notation

p is a prime, always ≥ 5 . ω is the Teichmüller character modulo p and $\langle x \rangle = x/\omega(x)$; following [18, Def. 8] we set $\zeta_p(s) = L_p(s, \omega^{1-s})$ for integers $s \geq 2$, the same normalisation as [8]. In particular $\zeta_p(2k) = 0$ for $k \geq 1$, used constantly. Γ_p is Morita's p -adic Gamma function [21], so that for a positive integer n

$$\Gamma_p(n) = (-1)^n \prod_{\substack{1 \leq j < n \\ p \nmid j}} j, \quad \text{and} \quad n! = p^{v_p(n!)} \prod_{i \geq 0} (-1)^{n_i+1} \Gamma_p(n_i+1), \quad n_i = \lfloor n/p^i \rfloor, \quad (1.3)$$

the product being the iterate of Morita's relation $\Gamma_p(n+1) = (-1)^{n+1} n! / (p^{\lfloor n/p \rfloor} \lfloor n/p \rfloor!)$ and finite because $\Gamma_p(1) = -1$ makes the terms with $n_i = 0$ equal to 1. We use throughout the

Beukers–Vlasenko expansion [8, eq. (2)]

$$\log \Gamma_p(y) = \Gamma'_p(0) y - \sum_{m \geq 2} \zeta_p(m) \frac{y^m}{m}, \quad y \in p\mathbb{Z}_p. \quad (1.4)$$

$\mathbb{D} := \mathbb{Z}[\mathbb{Z}_{>0}]$ is the free abelian group on the symbols $[x]$, $x \geq 1$, and $\mathcal{P} := \{(A, B) \in \mathbb{Z}^2 : A > B \geq 1\}$.

A notational warning, inherited from Paper C. The letters A, B do double duty: $(A(n), B(n))$ is the sporadic pair of *sequences* of §1.1, while in $c_p(A, B)$, $E_p(A, B)$ and $\delta(A, B)$ the arguments A, B are *positive integers*. The two never occur in the same formula, and the sequences are always written with their argument, $A(n)$, $B(n)$. We also set $c_p(A, A) := 1$, consistent with $\binom{Ap^t}{Ap^t} = 1$; the pairs (A, A) are excluded from $\mathcal{P}$ because they carry no information.

2 The Γ_p -tower

2.1 The tower product and the tower Beta

Lemma 2.1. [IMPORTED] (i)–(ii), [PROVED] (iii)–(iv). *Let p be odd.*

(i) $\Gamma_p(0) = 1$ and $|\Gamma_p(x) - \Gamma_p(y)|_p \leq |x - y|_p$ for all $x, y \in \mathbb{Z}_p$ ([21, 22]).

(ii) $\Gamma_p(1 + y) = -\Gamma_p(y)$ for $y \in p\mathbb{Z}_p$ (same sources).

(iii) Hence $\Gamma_p(y) \in 1 + p^e\mathbb{Z}_p$ and $v_p(\log \Gamma_p(y)) \geq e$ whenever $v_p(y) = e \geq 1$.

(iv) Hence $\Gamma'_p(0) \in \mathbb{Z}_p$.

Proof. (iii): by (i) with $x = 0$, $|\Gamma_p(y) - 1|_p \leq |y|_p = p^{-e}$; the logarithm of an element of $1 + p^e\mathbb{Z}_p$ lies in $p^e\mathbb{Z}_p$ for p odd. (iv): by (1.4), $\Gamma'_p(0) = \lim_{y \rightarrow 0} \log \Gamma_p(y)/y$ along $y \in p\mathbb{Z}_p$, and by (iii) the quotient lies in $\mathbb{Z}_p$ throughout. $\square$

Definition 2.2. For $x \in \mathbb{Z}_p$ put

$$\gamma_p(x) := \prod_{k \geq 1} \Gamma_p(xp^k) \in 1 + p\mathbb{Z}_p, \quad \widehat{g}_p(x) := - \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \zeta_p(m) \frac{x^m p^m}{m(1 - p^m)}.$$

We call γ_p the Γ_p -tower product and $\widehat{g}_p$ the Γ_p -tower function. For $B, D \geq 1$ the tower Beta is

$$\beta_p(B, D) := \frac{\gamma_p(B + D)}{\gamma_p(B) \gamma_p(D)}.$$

The product converges by Lemma 2.1(iii). For $\widehat{g}_p$ we use the following consequence of (1.4), and nothing else about ζ_p : since the series in (1.4) converges for *every* $y \in p\mathbb{Z}_p$, its coefficients satisfy

$$\varepsilon_m := v_p(\zeta_p(m)/m) \quad \text{obeys} \quad \varepsilon_m + m \longrightarrow \infty. \quad (2.1)$$

The m -th term of $\widehat{g}_p(x)$ then has valuation $\geq \varepsilon_m + m$ for $x \in \mathbb{Z}_p$, because $v_p(p^m/(1 - p^m)) = m$; so $\widehat{g}_p$ converges on $\mathbb{Z}_p$.

Proposition 2.3. [PROVED]. Write $\ell_p := \Gamma'_p(0) p/(1-p)$. Then for $x \in \mathbb{Z}_p$

$$\log \gamma_p(x) = \ell_p x + \hat{g}_p(x), \quad (2.2)$$

and consequently, for all $B, D \geq 1$,

$$\log \beta_p(B, D) = \hat{g}_p(B + D) - \hat{g}_p(B) - \hat{g}_p(D). \quad (2.3)$$

Moreover $\ell_p \in p\mathbb{Z}_p$ and $\hat{g}_p(x) \in p\mathbb{Z}_p$ for every $x \in \mathbb{Z}_p$, and γ_p satisfies the distribution relation

$$\gamma_p(px) = \frac{\gamma_p(x)}{\Gamma_p(px)}, \quad \text{equivalently} \quad \hat{g}_p(px) = \hat{g}_p(x) - \log(\Gamma_p(px) e^{-\Gamma'_p(0)px}). \quad (2.4)$$

Proof. Substituting $y = xp^k$ in (1.4) and summing over $k \geq 1$ requires interchanging the two summations. This is legitimate: by (2.1) the (k, m) term has valuation $\geq \varepsilon_m + mk \geq (\varepsilon_m + m) + 2(k-1)$, which tends to ∞ jointly in (k, m) , so the double family is summable and may be summed in either order. Using the p -adically convergent geometric series $\sum_{k \geq 1} p^k = p/(1-p)$ and $\sum_{k \geq 1} p^{km} = p^m/(1-p^m)$ this gives $\log \gamma_p(x) = \Gamma'_p(0)x p/(1-p) - \sum_{m \geq 2} \zeta_p(m)x^m p^m/(m(1-p^m))$; the even m drop out because $\zeta_p(2k) = 0$. This is (2.2). By Lemma 2.1(iii) each factor $\Gamma_p(xp^k)$ lies in $1 + p^k\mathbb{Z}_p$, so $\gamma_p(x) \in 1 + p\mathbb{Z}_p$ and $\log \gamma_p(x) \in p\mathbb{Z}_p$; and $\ell_p \in p\mathbb{Z}_p$ by Lemma 2.1(iv), whence $\hat{g}_p(x) = \log \gamma_p(x) - \ell_p x \in p\mathbb{Z}_p$. Since $(B+D) - B - D = 0$, the linear terms cancel in $\log \beta_p$, giving (2.3). Finally $\gamma_p(px) = \prod_{k \geq 1} \Gamma_p(xp^{k+1}) = \prod_{k \geq 2} \Gamma_p(xp^k) = \gamma_p(x)/\Gamma_p(px)$, and the second form follows on taking logarithms in (2.2). $\square$

Equation (2.3) is the point of the packaging: *the transcendental content of the tower depends on the pair only through the formal difference $[B+D] - [B] - [D]$* . This is what §3 exploits.

Remark 2.4 (relation to Beukers–Vlasenko). Write $G(y) = \Gamma_p(y)e^{-\Gamma'_p(0)y} = \exp(-\sum_{m \geq 2} \zeta_p(m)y^m/m)$, the generating function whose Taylor coefficients at 0 are the Frobenius structure constants α_j of [8]. Then $e^{\hat{g}_p(x)} = \prod_{k \geq 1} G(xp^k)$: the constants of this paper are *products of values* of G along a p -power tower, where the α_j are *Taylor coefficients* of G at 0. For the order-3 Apéry operator the available slots α_1, α_2 both vanish [3, §5.4], so no α_j can be one of our constants. This relationship was recorded as an orientation remark, explicitly disclaimed, in [2, §2.4]; Theorem 2.5 discharges it for the Kazandzidis limits.

2.2 The closed form

For $A > B \geq 1$ with $D = A - B$ set $A_l = \lfloor A/p^l \rfloor$, $B_l = \lfloor B/p^l \rfloor$, $D_l = \lfloor D/p^l \rfloor$ and

$$T(A, B) := \prod_{l \geq 1} \frac{(-1)^{A_l+1} \Gamma_p(A_l + 1)}{(-1)^{B_l+1} \Gamma_p(B_l + 1) \cdot (-1)^{D_l+1} \Gamma_p(D_l + 1)}, \quad E_p(A, B) := -p^{v_p(\frac{A}{B})} T(A, B) \frac{\Gamma_p(A+1)}{\Gamma_p(B+1) \Gamma_p(D+1)} \quad (2.5)$$

Both products are finite; $T(A, B) = 1$ when $A < p$. We call $E_p(A, B)$ the *elementary part*.

Theorem 2.5. [PROVED]. For every prime $p \geq 5$ and all integers $A > B \geq 1$ the limit $c_p(A, B) = \lim_t \left(\frac{A p^t}{B p^t} \right)$ exists in $\mathbb{Q}_p$, and

$$\boxed{c_p(A, B) = E_p(A, B) \cdot \beta_p(B, A - B).} \quad (2.6)$$

Moreover $v_p(c_p(A, B)) = v_p(\frac{A}{B})$, and the convergence is at 3 digits per level.

Proof. Fix $t \geq 1$ and apply (1.3) to Ap^t , Bp^t , Dp^t . By Kummer's theorem $v_p\left(\frac{Ap^t}{Bp^t}\right)$ counts the carries in adding Bp^t and Dp^t in base p , which is the carry count of $B+D$; hence $v_p\left(\frac{Ap^t}{Bp^t}\right) = v_p\left(\frac{A}{B}\right)$ for all t . The unit parts give

$$\left(\frac{Ap^t}{Bp^t}\right) = p^{v_p\left(\frac{A}{B}\right)} \prod_{i \geq 0} \frac{(-1)^{X_i+1} \Gamma_p(X_i+1)}{(-1)^{Y_i+1} \Gamma_p(Y_i+1) (-1)^{Z_i+1} \Gamma_p(Z_i+1)},$$

with $X_i = \lfloor Ap^t/p^i \rfloor$ and similarly for Y_i, Z_i . Split the product at $i = t$.

For $i > t$ one has $X_i = A_{i-t}$, $Y_i = B_{i-t}$, $Z_i = D_{i-t}$, so the tail is exactly $T(A, B)$, independent of t .

For $0 \leq i \leq t$ put $k = t - i$, so $X_i = Ap^k$, $Y_i = Bp^k$, $Z_i = Dp^k$. The signs contribute $(-1)^{Ap^k+1-(Bp^k+1)-(Dp^k+1)} = (-1)^{-1} = -1$ for each k , since $A = B + D$. For $k \geq 1$ the arguments lie in $p\mathbb{Z}_p$, where $\Gamma_p(1+y) = -\Gamma_p(y)$, so

$$\frac{\Gamma_p(Ap^k+1)}{\Gamma_p(Bp^k+1)\Gamma_p(Dp^k+1)} = \frac{-\Gamma_p(Ap^k)}{(-\Gamma_p(Bp^k))(-\Gamma_p(Dp^k))} = -\frac{\Gamma_p(Ap^k)}{\Gamma_p(Bp^k)\Gamma_p(Dp^k)},$$

and the two minus signs cancel. The factor $k = 0$ contributes $-\Gamma_p(A+1)/(\Gamma_p(B+1)\Gamma_p(D+1))$. Altogether

$$\left(\frac{Ap^t}{Bp^t}\right) = E_p(A, B) \prod_{k=1}^t \frac{\Gamma_p(Ap^k)}{\Gamma_p(Bp^k)\Gamma_p(Dp^k)}. \quad (2.7)$$

By Lemma 2.1 each factor lies in $1 + p^k\mathbb{Z}_p$ and differs from 1 by at most p^{-k} , so the partial products form a Cauchy sequence converging to $\gamma_p(A)/(\gamma_p(B)\gamma_p(D)) = \beta_p(B, D)$. Hence the limit (1.2) exists and equals (2.6). Since $E_p(A, B)$ is a p -power times a unit and β_p is a unit, $v_p(c_p) = v_p\left(\frac{A}{B}\right)$.

For the rate we quote the classical Jacobsthal–Kazandzidis congruence $\binom{np}{mp} \equiv \binom{n}{m} \pmod{p^{v_p(p^3nm(n-m))}}$ [16, 20]: at $n = Ap^t$, $m = Bp^t$ it gives

$$\left(\frac{Ap^{t+1}}{Bp^{t+1}}\right) \equiv \left(\frac{Ap^t}{Bp^t}\right) \pmod{p^{3+3t+v_p(ABD)}},$$

and since $v_p\left(\frac{Ap^t}{Bp^t}\right)$ is constant in t the relative gain is at least 3 digits per level. $\square$

Corollary 2.6. [PROVED]. $c_p(2, 1) = \lim_s \left(\frac{2p^s}{p^s}\right) = 2 \exp\left(-\sum_{m \geq 3 \text{ odd}} \zeta_p(m) \frac{(2^m - 2)p^m}{m(1 - p^m)}\right) = 2\beta_p(1, 1)$.

Proof. $T(2, 1) = 1$, $v_p\left(\frac{2}{1}\right) = 0$ and $\Gamma_p(3)/\Gamma_p(2)^2 = -2$, so $E_p(2, 1) = 2$; $\widehat{g}_p(2) - 2\widehat{g}_p(1)$ is the displayed series. $\square$

This recovers [3, Cor. 6.4]. The new content of Theorem 2.5 is not the formula but its shape: existence and closed form come from one computation, and the transcendental factor is a *single* tower Beta.

Finding 2.7. [VERIFIED: 11 pairs $(A, B) \times p \in \{5, 7, 11, 13\}$, all 12 certified digits, zero mismatches; at $p = 5$ also at 15 digits] Formula (2.6) was checked against $c_p(A, B)$ computed directly as a limit of exact binomial residues, for $(A, B) = (2, 1), (3, 1), (3, 2), (4, 1), (5, 2), (7, 3), (p, 1), (p, 2), (p+1, 1), (p+1, p), (2p, p)$ — including cases with carries, with $p \mid A$, and with $v_p(c_p) = 1$. The two sides are computed by disjoint code paths: the left from binomial residues via Legendre/Morita splitting, the right from the Kubota–Leopoldt values $\zeta_p(m)$.

2.3 Extension to $\mathbb{Z}[1/p]$

Proposition 2.8. [PROVED]. $c_p(pA, pB) = c_p(A, B)$ for all $A > B \geq 1$. Consequently c_p extends uniquely to pairs $A > B > 0$ in $\mathbb{Z}[1/p]$ by $c_p(A, B) := c_p(p^j A, p^j B)$ for any j with $p^j A, p^j B \in \mathbb{Z}$.

Proof. Directly from (1.2): $c_p(pA, pB) = \lim_t \binom{Ap^{t+1}}{Bp^{t+1}}$, a subsequence of the defining sequence for $c_p(A, B)$. $\square$

Proposition 2.8 is trivial from the definition but is a nontrivial identity for the closed form, and it is worth recording that (2.6) satisfies it on the nose.

Proposition 2.9. [PROVED]. Formula (2.6) is compatible with Proposition 2.8; the verification is the distribution relation (2.4).

Proof. $v_p\binom{pA}{pB} = v_p\binom{A}{B}$ (Kummer). Since $\lfloor pA/p^l \rfloor = A_{l-1}$,

$$T(pA, pB) = \frac{(-1)^{A+1} \Gamma_p(A+1)}{(-1)^{B+1} \Gamma_p(B+1) (-1)^{D+1} \Gamma_p(D+1)} T(A, B) = -T(A, B) \frac{\Gamma_p(A+1)}{\Gamma_p(B+1) \Gamma_p(D+1)},$$

using $A = B+D$ for the sign, while $\Gamma_p(pA+1)/(\Gamma_p(pB+1)\Gamma_p(pD+1)) = -\Gamma_p(pA)/(\Gamma_p(pB)\Gamma_p(pD))$ by $\Gamma_p(1+y) = -\Gamma_p(y)$. By (2.4), $\beta_p(pB, pD) = \beta_p(B, D) \Gamma_p(pB)\Gamma_p(pD)/\Gamma_p(pA)$. Multiplying the three displays, all Γ_p -factors at p -multiples and both extra signs cancel and one is left with $E_p(A, B)\beta_p(B, D)$. $\square$

Remark 2.10. The tower function itself does *not* descend: $\widehat{g}_p(px) \neq \widehat{g}_p(x)$, and (2.4) measures the discrepancy by a single Γ_p -value. It is the combination $E_p \cdot \beta_p$ that is p -homogeneous. This is the first sign that the natural invariants of the family are the tower Betas modulo Γ_p -values, which is exactly the quotient in which §3 works.

3 The relations

3.1 Three binomial relations

Proposition 3.1. [PROVED]. For c_p extended as in Proposition 2.8:

- (i) (boundary) $c_p(A, A) = 1$;
- (ii) (symmetry) $c_p(A, B) = c_p(A, A - B)$;
- (iii) (p -homogeneity) $c_p(pA, pB) = c_p(A, B)$;
- (iv) (trinomial revision) $c_p(A, B) c_p(B, C) = c_p(A, C) c_p(A - C, B - C)$ for $A > B > C \geq 1$.

Proof. Each is a binomial identity that holds at every level of the tower simultaneously, hence passes to the limit: $\binom{N}{K} = \binom{N}{N-K}$ for (ii), and $\binom{N}{K} \binom{K}{L} = \binom{N}{L} \binom{N-L}{K-L}$ at $N = Ap^t$, $K = Bp^t$, $L = Cp^t$ for (iv). (iii) is Proposition 2.8, (i) is trivial. For arguments in $\mathbb{Z}[1/p]$, multiply all of A, B, C by a common p^j clearing denominators and apply (iii). $\square$

Finding 3.2. [VERIFIED: $p \in \{5, 7, 11, 13\}$; 9 certified digits; 66 symmetry cells ($2 \leq A \leq 12$), 21 p -homogeneity cells ($2 \leq A \leq 7$) and 120 trinomial-revision cells ($3 \leq A \leq 10$) per prime; 828 checks in total, zero failures] Proposition 3.1 holds on the *directly computed* limits, i.e. without going through (2.6).

That these are verified as well as proved is not redundancy: the direct limits and the closed form are computed by disjoint code, and the relations are the only place where the two are compared across many pairs at once.

3.2 The divisor map

Definition 3.3. Define the group homomorphisms

$$\delta : \mathbb{Z}[\mathcal{P}] \longrightarrow \mathbb{D}, \quad (A, B) \longmapsto [A] - [B] - [A - B]; \quad m_r : \mathbb{D} \longrightarrow \mathbb{Z}, \quad \sum_x c_x [x] \longmapsto \sum_x c_x x^r \quad (r \geq 1);$$

$$\widehat{G} : \mathbb{D} \longrightarrow \mathbb{Q}_p, \quad \sum_x c_x [x] \longmapsto \sum_x c_x \widehat{g}_p(x).$$

By (2.3), for any $\mathbf{n} = \sum_i n_i (A_i, B_i) \in \mathbb{Z}[\mathcal{P}]$,

$$\prod_i \beta_p(B_i, A_i - B_i)^{n_i} = \exp(\widehat{G}(\delta(\mathbf{n}))). \quad (3.1)$$

Everything in this section is a consequence of (3.1).

Proposition 3.4. [PROVED]. $\text{im } \delta = \ker m_1$, a subgroup of $\mathbb{D}$ of infinite rank and corank 1.

Proof. $m_1 \delta(A, B) = A - B - (A - B) = 0$, so $\text{im } \delta \subseteq \ker m_1$. Conversely $\delta(2, 1) = [2] - 2[1]$, and $\delta(n, 1) = [n] - [1] - [n - 1]$ gives inductively $[n] - n[1] = \delta(n, 1) + ([n - 1] - (n - 1)[1]) \in \text{im } \delta$ for all $n \geq 2$. Finally any $\mu = \sum_x c_x [x]$ with $m_1(\mu) = 0$ equals $\sum_x c_x ([x] - x[1])$, hence lies in $\text{im } \delta$. $\square$

Lemma 3.5. [PROVED]. The map $\mu \mapsto (m_3(\mu), m_5(\mu), m_7(\mu), \dots)$ is injective on $\mathbb{D}$. Indeed if $m_r(\mu) = 0$ for infinitely many r then $\mu = 0$.

Proof. Suppose $\mu = \sum_x c_x [x] \neq 0$ and let X be the largest x with $c_x \neq 0$. Then $m_r(\mu)/X^r = c_X + \sum_{x < X} c_x (x/X)^r \rightarrow c_X \neq 0$ as $r \rightarrow \infty$ in $\mathbb{R}$, so $m_r(\mu) \neq 0$ for all large r . $\square$

3.3 The rigidity theorem

Theorem 3.6 (Rigidity). [PROVED]. Let $\mathbf{n} = \sum_{i=1}^r n_i (A_i, B_i) \in \mathbb{Z}[\mathcal{P}]$ and put $\mu = \delta(\mathbf{n}) \in \mathbb{D}$, $v_m = m_m(\mu) \in \mathbb{Z}$.

(i) If $\mu = 0$ then, for every prime $p \geq 5$,

$$\prod_i \beta_p(B_i, A_i - B_i)^{n_i} = 1, \quad \text{hence} \quad \prod_i c_p(A_i, B_i)^{n_i} = \prod_i E_p(A_i, B_i)^{n_i},$$

an explicit rational number times a monomial in Γ_p -values at positive integers.

(ii) If $\mu \neq 0$ and $\prod_i \beta_p(B_i, A_i - B_i)^{n_i} = 1$ for some $p \geq 5$, then

$$\sum_{\substack{m \geq 3 \\ m \text{ odd}}} v_m \zeta_p(m) \frac{p^m}{m(1-p^m)} = 0 \quad (3.2)$$

is a nontrivial vanishing p -adic linear form in the odd Kubota–Leopoldt zeta values: the integers v_m are not all zero, and $|v_m| \leq (\sum_x |c_x|) X^m$ where X is the largest element of the support of μ .

Proof. (i) is (3.1) with $\widehat{G}(0) = 0$, together with Theorem 2.5. For (ii): by Proposition 2.3, $\widehat{G}(\mu) \in p\mathbb{Z}_p$, and $\exp$ is injective on $p\mathbb{Z}_p$ for p odd, so (3.1) gives $\widehat{G}(\mu) = 0$. Each $\widehat{g}_p(x)$ is by Definition 2.2 a convergent series in $\mathbb{Q}_p$, and μ has finite support, so the finite sum $\sum_x c_x \widehat{g}_p(x)$ may be resummed term by term:

$$\widehat{G}(\mu) = - \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \left(\sum_x c_x x^m \right) \zeta_p(m) \frac{p^m}{m(1-p^m)} = - \sum_{\substack{m \geq 3 \\ m \text{ odd}}} v_m \zeta_p(m) \frac{p^m}{m(1-p^m)}.$$

Lemma 3.5 says the v_m are not all zero, and $|v_m| = |\sum_x c_x x^m| \leq (\sum_x |c_x|) X^m$. $\square$

Hypothesis 3.7 (IND_p). [OPEN]. The only $\mu \in \mathbb{D}$ with $\widehat{G}(\mu) = 0$ is $\mu = 0$; equivalently, the values $\widehat{g}_p(1), \widehat{g}_p(2), \widehat{g}_p(3), \dots$ are $\mathbb{Q}$ -linearly independent; equivalently, (3.2) has no solution in integers v_m of the form $v_m = m_m(\mu)$, $\mu \neq 0$.

Corollary 3.8. [PROVED], under Hypothesis 3.7. The multiplicative group generated by $\{\beta_p(B, D) : B, D \geq 1\}$ inside $1 + p\mathbb{Z}_p$ is free abelian of countably infinite rank, and δ induces an isomorphism onto $\ker m_1 \subset \mathbb{D}$. Unconditionally it is a quotient of $\ker m_1$.

Proof. By (3.1) the group is the image of $\mathbb{Z}[\mathcal{P}]$ under $\exp \circ \widehat{G} \circ \delta$. Its relation module contains $\ker \delta$ always, and equals $\ker \delta$ exactly when $\ker \widehat{G} \cap \text{im } \delta = 0$, which is Hypothesis 3.7. Then the group is $\mathbb{Z}[\mathcal{P}] / \ker \delta \cong \text{im } \delta = \ker m_1$ by Proposition 3.4. Finally $\ker m_1$ is free, being a subgroup of the free abelian group $\mathbb{D}$, and of infinite rank, since it contains the independent family $\{[n] - n[1]\}_{n \geq 2}$. $\square$

3.4 The lattice of binomial relations

Theorem 3.6(i) makes $\ker \delta$ the lattice of *expected* relations. It contains two visible families, coming from Proposition 3.1(ii) and (iv):

$$(S_{A,B}) : (A, B) - (A, A - B), \quad (3.3)$$

$$(T_{A,B,C}) : (A, B) + (B, C) - (A, C) - (A - C, B - C), \quad A > B > C \geq 1. \quad (3.4)$$

Both are in $\ker \delta$: for (3.4), $\delta(A, B) + \delta(B, C) = [A] - [A - B] - [C] - [B - C] = \delta(A, C) + \delta(A - C, B - C)$. Note that p -homogeneity is *not* of this type — $\delta(pA, pB) \neq \delta(A, B)$ — consistently with the fact that it moves β_p by a Γ_p -value (Proposition 2.9) rather than fixing it.

Finding 3.9. [VERIFIED: $N \leq 11$; exact integer linear algebra, Smith normal forms over $\mathbb{Z}$] Let $\mathcal{P}_N = \{(A, B) : N \geq A > B \geq 1\}$ and δ_N the restriction of δ . Then

$$\text{rank } \delta_N = N - 1, \quad \text{rank } \ker \delta_N = \binom{N}{2} - (N - 1) = \binom{N - 1}{2},$$

and $\ker \delta_N$ is generated *over* $\mathbb{Z}$ by the relations (3.3) and (3.4) with all indices $\leq N$. The argument is: the sublattice L_N they span is contained in $\ker \delta_N$ (checked symbolically, not assumed); it has the same rank, hence the same $\mathbb{Q}$ -span; and the Smith normal forms of both have all elementary divisors equal to 1, so both are saturated in $\mathbb{Z}[\mathcal{P}_N]$. Two saturated sublattices with the same $\mathbb{Q}$ -span coincide.

The first assertion is Proposition 3.4 restricted, hence [PROVED]; the generation statement is [VERIFIED:] only.

Problem 3.10. [OPEN]. Prove that $\ker \delta$ is generated by (3.3) and (3.4). Equivalently: give a presentation of the free abelian group $\ker m_1$ by the symbols (A, B) and those two families of relations.

3.5 What the theorem says, in words

Combining: the tower Betas satisfy the binomial relations (3.3), (3.4) — and, under Hypothesis 3.7, *nothing else*. Equivalently, at the level of the Kazandzidis limits: the only relations

$$\prod_i c_p(A_i, B_i)^{n_i} = \prod_i E_p(A_i, B_i)^{n_i}$$

— those in which the transcendental factor cancels *identically* — are the binomial ones. A single unexpected multiplicative relation among the tower Betas would, by Theorem 3.6(ii), produce a nontrivial vanishing $\mathbb{Q}$ -linear form in $\zeta_p(3), \zeta_p(5), \zeta_p(7), \dots$ with explicit integer coefficients of controlled growth. Since no odd p -adic zeta value is known to be irrational for any $p \geq 5$, that would be a substantial result about p -adic L -values. The relation problem for the building blocks of the Λ -algebra therefore sits directly on top of the p -adic odd-zeta problem, and not beside it.

Remark 3.11 (what is *not* claimed). Theorem 3.6 does not determine when $\prod_i c_p(A_i, B_i)^{n_i}$ merely *lies in* $\mathcal{E}_p$. That would require knowing $\mathcal{E}_p \cap (1 + p\mathbb{Z}_p)$, which contains $\mathbb{Q}^\times \cap (1 + p\mathbb{Z}_p)$ and is therefore large; the transcendental factor can in principle be absorbed by an elementary one without vanishing. Everything we prove is about the tower Betas, where the elementary part has been divided out by construction.

4 The Λ -algebra

4.1 Definition

Definition 4.1. Fix $p \geq 5$. Let

$$\begin{aligned} \mathcal{E}_p &:= \text{the subgroup of } \mathbb{Q}_p^\times \text{ generated by } \mathbb{Q}^\times \text{ and } \{\Gamma_p(n) : n \in \mathbb{Z}_{\geq 1}\}, & (\text{the elementary group}) \\ \mathcal{K}_p &:= \text{the } \mathbb{Q}\text{-subalgebra of } \mathbb{Q}_p \text{ generated by } \mathcal{E}_p \cup \{\beta_p(B, D) : B, D \geq 1\}, & (\text{the Kazandzidis algebra}) \\ \mathcal{L}_p &:= \text{the closure of } \mathcal{K}_p[h_p^{(w)}(a) : a \geq 1, w \geq 2] \text{ in } \mathbb{Q}_p, & (\text{the } \Lambda\text{-algebra}) \end{aligned}$$

where

$$h_p^{(w)}(a) := \lim_{s \rightarrow \infty} p^{ws} \sum_{k \leq ap^s} k^{-w} = \sum_{\substack{x \in \mathbb{Z}[1/p] \\ 0 < x \leq a}} x^{-w}$$

is the *tower-harmonic constant*. For $w = 3$ this is [3, Thm. 6.1]; the proof there — split $\sum_{k \leq n} k^{-w}$ at $p \mid k$, iterate, and re-index $x = k/p^i$ — is verbatim the same for every $w \geq 2$, and we use it in that generality. We write $h_p = h_p^{(3)}$.

By Theorem 2.5, $c_p(A, B) \in \mathcal{K}_p$ for all $A > B \geq 1$. $\mathcal{L}_p$ is a closed $\mathbb{Q}$ -subalgebra of $\mathbb{Q}_p$; it contains a multiplicative group of infinite rank (Proposition 4.12 under Hypothesis 3.7).

It is worth saying which Γ_p -values are and are not in play. The Γ_p -values generating $\mathcal{E}_p$ are at *positive integers*, where Γ_p is an explicit product of units by (1.3), and at p -power arguments, where it is the tower product γ_p . The Γ_p -values at rational arguments with small denominator, by contrast, are *algebraic*: $\Gamma_p(1/2)^2 = \pm 1$ by reflection and $\Gamma_p(1/3), \Gamma_p(1/4)$ are Gauss-sum values by Gross–Koblitz [15]. Those contribute nothing new to $\mathcal{L}_p$, and are not where the transcendence of these constants can live; the tower is.

Proposition 4.2. [PROVED]. $h_p^{(w)}(pa) = p^{-w}h_p^{(w)}(a)$ for every $a > 0$ in $\mathbb{Z}[1/p]$, and $h_p^{(w)}$ extends to $\mathbb{Z}[1/p]_{>0}$ by this relation.

Proof. $x \mapsto x/p$ is a bijection from $\{x \in \mathbb{Z}[1/p] : 0 < x \leq pa\}$ onto $\{y \in \mathbb{Z}[1/p] : 0 < y \leq a\}$, and $x^{-w} = p^{-w}y^{-w}$. $\square$

This is the tower-harmonic shadow of the scaling law $\Lambda_{p^k a} = \chi(p)^k p^{-wk} \Lambda_a$ of [3, Prop. 2.5(iii)].

Remark 4.3 (the index range, which matters). Paper C’s series for $h_p^{(w)}$ reads $h_p^{(w)}(a) = \sum_i p^{wi} S_i(a)$, $S_i(a) = \sum_{k \leq ap^i, p \nmid k} k^{-w}$, with i running over *all* integers with $ap^i \geq 1$. For $1 \leq a < p$ this is $i \geq 0$ and the two descriptions of $h_p^{(w)}(a)$ agree term by term; for $a > p$ the terms with $i < 0$ are exactly the x in the sum with $v_p(x) > 0$, and omitting them breaks Proposition 4.2. We record it because Proposition 4.2 evaluates $h_p^{(w)}$ outside the range $1 \leq a < p$ in which the two descriptions are visibly the same, and because the omission is an easy one to make: an implementation restricted to $i \geq 0$ passes every test at $a < p$ and fails Proposition 4.2 at $a = p$, which is how Finding 4.4(a) was used here.

Finding 4.4. [VERIFIED: $p \in \{5, 7\}$, $w = 3$; $a \leq 5$; at 12 and again at 18 certified p -adic digits] (a) Proposition 4.2 holds numerically at $a = 1, 2, 3$. (b) No $\mathbb{Q}$ -affine relation among $h_p(1), \dots, h_p(N)$, $N \leq 5$, was found below the noise floor; every best height sits at the floor and scales with the precision — e.g. at $p = 5$, $N = 1$ the best height moves from 1.6×10^4 (floor 1.6×10^4) to 1.6×10^6 (floor 2.0×10^6) when the precision goes from 12 to 18 digits, and at $p = 7$, $N = 3$ from 1.7×10^2 (floor 3.4×10^2) to 4.0×10^3 (floor 6.4×10^3). Paper C excludes $h_p(1) \in \mathbb{Q}$ and $h_p(1)$ quadratic over $\mathbb{Q}$ [3, §5.2]; the relations *among* different tower bases are new here.

4.2 The structure theorem, proved

Paper C establishes the block decomposition of Λ_a for $a = 1$ block by block, as a Finding. Here we prove it, for every tower base. Throughout this subsection (a_n, b_n) is the classical Apéry $\zeta(3)$ pair, $a_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and b_n the second solution with $b_0 = 0$, $b_1 = 6$, so $w = 3$ and $\chi = 1$.

Lemma 4.5. [PROVED]. Fix $p \nmid a$ and set $\kappa = \kappa(a, p) := \lfloor \log_p(2a) \rfloor + 2$.

(i) For all $N \geq K \geq 0$, $v_p\left(\frac{N}{K}\right) \leq \lfloor \log_p N \rfloor + 1$.

- (ii) For $i \geq 0$ and $1 \leq j \leq ap^i$ with $p \nmid j$, $i \leq v_p(ap^i_j) \leq i + \kappa - 1$ and $v_p(ap^{i+j}_j) \leq i + \kappa - 1$.
- (iii) $v_p(ap^{s-i}_{jp^{s-i}}) = v_p(ap^i_j)$ and $v_p(ap^{s+jp^{s-i}}_{jp^{s-i}}) = v_p(ap^{i+j}_j)$ for every $s \geq i$.

Proof. By Kummer's theorem [14, §1] $v_p(\binom{N}{K})$ is the number of carries in adding K and $N - K$ in base p . (i): all carries occur in positions $0, \dots, \lfloor \log_p N \rfloor$, since N has $\lfloor \log_p N \rfloor + 1$ digits. (iii): multiplying N and K by p^{s-i} shifts all digits and changes no carry; note $ap^s + jp^{s-i} = (ap^i + j)p^{s-i}$. (ii), lower bound: take $N = ap^i$, $K = j$; the last i base- p digits of N are 0 while K has a nonzero units digit, so a carry occurs out of position 0 and, inductively, out of every position $< i$. Upper bounds: apply (i) with $N = ap^i$ and with $N = ap^i + j \leq 2ap^i$, both of which are $< 2ap^i \cdot p$, so $\lfloor \log_p N \rfloor + 1 \leq i + \lfloor \log_p(2a) \rfloor + 1 = i + \kappa - 1$. $\square$

Theorem 4.6. [PROVED]. Let $p \geq 5$ and $p \nmid a$. The limit $\tilde{A}_a := \lim_s a_{ap^s}$ exists in $\mathbb{Z}_p$ and

$$\tilde{A}_a = 1 + \sum_{i \geq 0} \sum_{\substack{1 \leq j \leq ap^i \\ p \nmid j}} \left[c_p(ap^i, j) c_p(ap^i + j, j) \right]^2, \quad (4.1)$$

with the double series convergent in $\mathbb{Z}_p$. We write

$$u(i, j) := \left[c_p(ap^i, j) c_p(ap^i + j, j) \right]^2, \quad v_p(u(i, j)) \geq 2i, \quad (4.2)$$

for the (i, j) block. (For $i = 0$, $j = a$ the first factor is $c_p(a, a) = 1$.)

Proof. Fix s and split the sum $a_{ap^s} = \sum_{k=0}^{ap^s} A(ap^s, k)$, $A(n, k) = \binom{n}{k}^2 \binom{n+k}{k}^2$, according to $k = 0$ and $k = jp^{s-i}$ with $p \nmid j$, $i = s - v_p(k) \in [0, s]$ and $1 \leq j \leq ap^i$. This is a bijection between $\{1, \dots, ap^s\}$ and $\{(i, j) : 0 \leq i \leq s, 1 \leq j \leq ap^i, p \nmid j\}$, and $A(ap^s, 0) = 1$.

Write $u_s(i, j) := A(ap^s, jp^{s-i})$. By Lemma 4.5, $v_p(ap^{s-i}_j) = v_p(ap^i_j) \geq i$, so

$$v_p(u_s(i, j)) \geq 2i \quad \text{for all } s \geq i, \quad (4.3)$$

uniformly. Moreover, since $ap^s = (ap^i)p^{s-i}$ and $ap^s + jp^{s-i} = (ap^i + j)p^{s-i}$, Theorem 2.5 gives

$$\lim_{s \rightarrow \infty} u_s(i, j) = \left[c_p(ap^i, j) c_p(ap^i + j, j) \right]^2 = u(i, j), \quad (4.4)$$

and $v_p(u(i, j)) \geq 2i$ by the same bound, which is (4.2).

Now fix $I \geq 1$. For every $s \geq I$, by (4.3),

$$\left| a_{ap^s} - 1 - \sum_{i < I} \sum_j u_s(i, j) \right|_p \leq p^{-2I}, \quad \left| \sum_{i \geq I} \sum_j u(i, j) \right|_p \leq p^{-2I},$$

the second because a p -adic series with terms of valuation $\geq 2I$ converges and has valuation $\geq 2I$. The inner sum $\sum_{i < I} \sum_j$ is finite, so by (4.4) there is $S(I)$ with $|\sum_{i < I} \sum_j (u_s(i, j) - u(i, j))|_p \leq p^{-2I}$ for $s \geq S(I)$. Combining the three estimates, $|a_{ap^s} - \Sigma|_p \leq p^{-2I}$ for $s \geq \max(I, S(I))$, where Σ denotes the right-hand side of (4.1) (which converges, again by the valuation bound). As I was arbitrary, $a_{ap^s} \rightarrow \Sigma$. $\square$

Remark 4.7. The proof does not use the Beukers–Coster supercongruence $a_{mp^r} \equiv a_{mp^{r-1}} \pmod{p^{3r}}$ [24]; existence of $\tilde{A}_a$ falls out of the block estimate. It does not recover the p^{3r} either: (4.3) gives only 2 digits per level, and the sharp rate is a genuinely finer fact.

Theorem 4.8. [PROVED]. Let $p \geq 5$ and $p \nmid a$. Then $\tilde{B}_a := \lim_s p^{3s} b_{ap^s}$ exists and

$$\tilde{B}_a = h_p(a) \tilde{A}_a + \sum_{(i,j)} \sum_{(i',j')} (-1)^{j'-1} \frac{u(i,j) p^{3i'}}{2 j'^3 c_p(ap^{i'}, j') c_p(ap^{i'} + j', j')}, \quad (4.5)$$

the inner sum being restricted to the pairs with $j' p^{-i'} \leq j p^{-i}$, with the $((i,j), (i',j'))$ block of valuation $\geq 2i + i' - 2\kappa + 2$, with $\kappa = \kappa(a, p)$ as in Lemma 4.5, and both series convergent. In particular, if $\tilde{A}_a \neq 0$ then

$$\Lambda_a = \frac{\tilde{B}_a}{\tilde{A}_a} = h_p(a) + \frac{1}{\tilde{A}_a} \sum_{(i,j), (i',j')} (\cdots) \in \mathcal{L}_p. \quad (4.6)$$

Proof. Apéry's formula [5, 12] reads $b_n = \sum_k A(n, k) [H_3(n) + T(n, k)]$ with $H_3(n) = \sum_{k \leq n} k^{-3}$ and $T(n, k) = \sum_{m \leq k} (-1)^{m-1} (2m^3 \binom{n}{m} \binom{n+m}{m})^{-1}$. Hence

$$p^{3s} b_{ap^s} = (p^{3s} H_3(ap^s)) a_{ap^s} + \sum_k A(ap^s, k) \sum_{m \leq k} \frac{(-1)^{m-1} p^{3s}}{2m^3 \binom{ap^s}{m} \binom{ap^s+m}{m}}.$$

The first summand converges to $h_p(a) \tilde{A}_a$ by [3, Thm. 6.1] and Theorem 4.6. For the second, index $k = j p^{s-i}$ and $m = j' p^{s-i'}$ as before. Then $v_p(m^3) = 3(s - i')$ and, by Lemma 4.5(ii)–(iii), $v_p\left(\frac{ap^s}{m}\right) = v_p\left(\frac{ap^{i'}}{j'}\right) \leq i' + \kappa - 1$ and likewise $v_p\left(\frac{ap^s+m}{m}\right) = v_p\left(\frac{ap^{i'}+j'}{j'}\right) \leq i' + \kappa - 1$, so the inner term

$$w_s(i', j') := \frac{p^{3s}}{2m^3 \binom{ap^s}{m} \binom{ap^s+m}{m}} \quad \text{has} \quad v_p(w_s(i', j')) \geq i' - 2\kappa + 2,$$

uniformly in s , and by Theorem 2.5 $w_s(i', j') \rightarrow p^{3i'} (2j'^3 c_p(ap^{i'}, j') c_p(ap^{i'} + j', j'))^{-1}$. Together with (4.3) the $((i,j), (i',j'))$ block has valuation $\geq 2i + i' - 2\kappa + 2$, which tends to ∞ jointly in (i, i') ; the same three-estimate argument as in Theorem 4.6, now with a double truncation, gives (4.5). Finally every ingredient of (4.5) lies in $\mathcal{K}_p[h_p(a)]$ and the series converges in $\mathbb{Q}_p$, so $\tilde{B}_a$ and, when $\tilde{A}_a \neq 0$, Λ_a lie in $\mathcal{L}_p$. $\square$

Remark 4.9 (on $\tilde{A}_a \neq 0$). Λ_a exists unconditionally by [3, Thm. 2.4], so $\Lambda_a \tilde{A}_a = \tilde{B}_a$ holds whenever $\tilde{A}_a \neq 0$, and $\tilde{A}_a \neq 0$ as soon as $v_p(\tilde{A}_a) < \infty$. [VERIFIED: $p \in \{5, 7\}$, $a \in \{1, 2\}$: $v_p(\tilde{A}_a) \in \{0, 1\}$; and $v_p(\tilde{A}_1) \leq 1$ at $p = 5, 7, 11, 13$ [3, Tab. 16]], so the hypothesis is harmless in the computed range; we do not know it in general. This is the only conditional step in (4.6).

Corollary 4.10. [PROVED]. Let $p \geq 5$, $p \nmid a$, and assume $\tilde{A}_a \neq 0$ (see Remark 4.9: this is not known in general, and is [VERIFIED:] only on the cells listed there). Then, for the classical Apéry pair, $\Lambda_a \in \mathcal{L}_p$.

Finding 4.11. [VERIFIED: $p \in \{5, 7\}$; $a = 1, 2$; 12 certified digits; $\tilde{A}_a$ computed independently from the exact integer double sum at $n = ap^s$, $s \leq 4$ resp. 3] Truncating (4.1) at $i \leq I$ leaves a residual of valuation 3–4 for $I = 0$ and 6–7 for $I = 1$, against the guaranteed $2(I + 1)$; every predicted block is confirmed and the residual moves to the next block. Paper C records the case $a = 1$; the case $a = 2$ is new here and is the numerical companion to the general- a statement of Theorem 4.6.

4.3 Why no finite set of tower Betas suffices

Proposition 4.12. [PROVED], under Hypothesis 3.7. For every $N \geq 2$ the group generated by $\{\beta_p(B, D) : B + D \leq N\}$ is free abelian of rank exactly $N - 1$, and the lattice of relations among the images of the $\binom{N}{2}$ symbols of $\mathcal{P}_N$ has rank $\binom{N-1}{2}$. Consequently no finitely generated subgroup of $1 + p\mathbb{Z}_p$ contains all the tower Betas.

Proof. The generators are indexed by $\mathcal{P}_N$, and by (3.1) the group is the image of $\mathbb{Z}[\mathcal{P}_N]$ under $\exp \circ \widehat{G} \circ \delta$. Under Hypothesis 3.7 the kernel of that map is $\ker \delta_N$, so the group is isomorphic to $\text{im } \delta_N$, of rank $N - 1$ by Proposition 3.4; the relation lattice then has rank $\binom{N}{2} - (N - 1) = \binom{N-1}{2}$. The ranks are unbounded in N , so no finitely generated subgroup contains all the β_p . $\square$

Remark 4.13. We do *not* deduce that $\mathcal{K}_p$ is not a finitely generated $\mathbb{Q}$ -algebra: a finitely generated algebra can perfectly well contain a multiplicative group of infinite rank (as $\mathbb{Q}$ does). The correct statement is the one about groups, and it is the one the asymmetry of §5 uses.

The decomposition (4.1) uses $c_p(ap^i, j)$ for all i , hence tower Betas at unboundedly large arguments. Under Hypothesis 3.7 these are independent, so the p -adic Apéry limit is not merely *an* infinite sum: it is an infinite sum over a basis. This is the precise sense in which the one-generator hypothesis refuted numerically in [3, §6.6] fails structurally.

5 The mirror

5.1 What the two completions share

Theorem 5.1 (Transfer Principle; Paper D [4]). [IMPORTED]. Let $R \in \mathbb{Q}(t)$ have poles only at non-positive integers and $\deg R \leq -2$, with partial fractions $R(t) = \sum_{i,k} r_{i,k}(t+k)^{-i}$; set $\rho_i = \sum_k r_{i,k}$ and $\rho_{0,\theta} = -\sum_{i,k} r_{i,k} \sum_{\nu=0}^{k-1} (\nu+\theta)^{-i}$. Then

$$\sum_{m \geq 0} R(m+\theta) = \rho_{0,\theta} + \sum_{i \geq 2} \rho_i \zeta(i, \theta), \quad - \int_{\mathbb{Z}_p} \widetilde{R}(t+\theta) dt = \rho_{0,\theta} + \sum_{i \geq 2} \rho_i \omega(\theta)^{1-i} \zeta_p(i, \theta),$$

with the same rational numbers ρ , $\widetilde{R}$ a primitive of R . Consequently every transfer identity and every integrality lemma of Rhin–Viola type, being a statement about the ρ ’s, holds verbatim in both completions.

The Transfer Principle says that the *rational skeleton* of an Apéry-type construction — which linear form in which zeta values, with which coefficients — is completion-independent. Nothing in the construction knows which completion it is in until the values $\zeta(i, \theta)$ or $\zeta_p(i, \theta)$ are attached.

5.2 What they do not share

Against that shared skeleton, the constants produced are of wholly different kinds.

Proposition 5.2 (archimedean tameness). [IMPORTED]. For each of the twelve sporadic pairs whose characteristic roots have distinct moduli, the archimedean Apéry limit $\lim_n B(n)/A(n)$ exists and is a rational multiple of $\zeta(w)$ or of $L(\chi_D, w)$ with $w \in \{2, 3\}$ [25, 6, 10, 1]. In particular each spans a one-dimensional $\mathbb{Q}$ -vector space, and the $\mathbb{Q}$ -span of all twelve is contained in the

five-dimensional space $\mathbb{Q}\zeta(2) + \mathbb{Q}\zeta(3) + \mathbb{Q}L(\chi_{-3}, 2) + \mathbb{Q}L(\chi_{-4}, 2) + \mathbb{Q}L(\chi_{-3}, 3)$. (That the five are $\mathbb{Q}$ -linearly independent is of course not known; five is an upper bound for the dimension, which is all we use.)

Theorem 5.3 (the asymmetry). [PROVED], under Hypothesis 3.7, for the classical Apéry pair. Fix $p \geq 5$ and $p \nmid a$. The block decomposition (4.6) of the single constant Λ_a involves the tower Betas $\beta_p(j, ap^i - j)$ and $\beta_p(j, ap^i)$ for every $i \geq 0$ and every $j \leq ap^i$ prime to p . The group Γ_a they generate inside $1 + p\mathbb{Z}_p$ is free abelian of infinite rank.

Proof. Theorem 4.8 exhibits the blocks. By (3.1) and Hypothesis 3.7, Γ_a is isomorphic to the image under δ of the free abelian group on those pairs; that image contains $\mu_i := \delta(ap^i, 1) = [ap^i] - [1] - [ap^i - 1]$ for every $i \geq 0$. The largest symbol occurring in μ_i is $[ap^i]$, with coefficient $+1$, and the ap^i are strictly increasing; so in any non-trivial finite $\mathbb{Z}$ -combination $\sum n_i \mu_i$ the symbol $[ap^{i_{\max}}]$ with $i_{\max}$ the largest index with $n_i \neq 0$ survives with coefficient $n_{i_{\max}} \neq 0$. Hence the μ_i are $\mathbb{Z}$ -independent and the image has infinite rank (Proposition 3.4, Corollary 3.8). $\square$

The contrast is therefore not a matter of how hard the two sides are to compute. The archimedean limit is a value of one period at one singular point; the p -adic limit is a sum over the whole p -adic tower tree, and it collects the generating function G at *every* level. A one-dimensional space on one side, an infinite-rank one on the other, with the rational skeleton identical. Only the values attached differ, and ζ_p is attached infinitely often where ζ is attached once.

Remark 5.4 (the limitless trio). Three of the fifteen families — B , (δ) , (η) — have characteristic roots that are complex conjugates of equal modulus and hence *no* archimedean Apéry limit at all, yet all three have p -adic tower limits: (δ) outright, B and (η) after the $\chi(p)^{-s}$ twist [3, §7], the sole exception being (η) at the ramified prime $p = 5$. Proposition 5.2 is thus vacuous for them while Theorem 5.3's p -adic side is not: on those three families the asymmetry is total. (The observation that the p -adic limit exists where the archimedean one does not is due to [1, §8.2]; what is added here is the algebra it lives in.)

6 The exclusion ledger

Everything conditional above rests on Hypothesis 3.7. We cannot prove it; we can and do report what a search finds.

6.1 Instrument and convention

Reference values $\zeta_p(m)$, for all odd m up to a cutoff a few beyond the target precision (the m -th term of $\widehat{g}_p$ has valuation $\geq \varepsilon_m + m$ by (2.1)), come from the Kubota–Leopoldt implementation of [3, App. A], built from a Volkenborn series and validated on 980 independent interpolation identities, on independence of the auxiliary modulus, and against the limit characterisation of [18, 8]. The search instrument is an exact-Fraction LLL with the residue column weighted by p^A , so that only exact relations modulo p^A survive. For an m -term relation certified to A digits the heuristic noise floor is $p^{A/m}$: a lattice of that determinant contains vectors of that size for accidental reasons. LLL does not *exclude* relations: it fails to find them, with an approximation guarantee. What we therefore report is: no vector below the floor was returned, and LLL returns a vector within a factor $2^{(n-1)/2}$ of the shortest, so a genuine relation substantially below the

floor would have been returned. The decisive test is the second one: *a genuine relation's height does not move when A increases*, whereas noise tracks the floor. All of this is evidence, not exclusion.

Finding 6.1. [VERIFIED: $p \in \{5, 7, 11, 13\}$; at the same precision as the searches below] Adjoin to the basis the number $y := 3\hat{g}_p(1) - 7\hat{g}_p(2)$, computed from the same data. The finder recovers the planted relation $3\hat{g}_p(1) - 7\hat{g}_p(2) - y = 0$, of height 7, at every prime tested.

6.2 The search

Finding 6.2 (nothing below the floor). [VERIFIED: $p \in \{5, 7, 11, 13\}$; $x = 1, \dots, 8$; at $A = 26$ and again at $A = 38$ certified p -adic digits] No $\mathbb{Q}$ -linear relation among $\hat{g}_p(1), \dots, \hat{g}_p(N)$, and no $\mathbb{Q}$ -affine one (i.e. with 1 adjoined), was found below the noise floor, for any $N \leq 8$. Every best height sits at the floor and *scales with the precision*, which is the signature of noise. Table 1 records the two-precision comparison at $p = 5$ and $p = 7$; the other two primes behave identically.

Table 1: (IND_p) : best LLL height found against the noise floor $p^{A/m}$, at two certified precisions. A genuine relation would have a height that does not move between the two columns.

basis	$p = 5$		$p = 7$	
	$A = 26$ height / floor	$A = 38$ height / floor	$A = 26$ height / floor	$A = 38$ height / floor
$\hat{g}_p(1), \hat{g}_p(2)$	8.6e8 / 1.2e9	1.9e13 / 1.9e13	7.4e10 / 9.7e10	1.0e15 / 1.1e16
$\hat{g}_p(1..3)$	3.1e5 / 1.1e6	3.8e8 / 7.1e8	8.1e6 / 2.1e7	2.2e10 / 5.1e10
$\hat{g}_p(1..4)$	1.1e4 / 3.5e4	2.3e6 / 4.4e6	1.6e5 / 3.1e5	7.4e7 / 1.1e8
$\hat{g}_p(1..6)$	4.5e2 / 1.1e3	1.5e4 / 2.7e4	2.7e3 / 4.6e3	8.9e4 / 2.3e5
$\hat{g}_p(1..8)$	95 / 1.9e2	9.2e2 / 2.1e3	2.5e2 / 5.6e2	4.0e3 / 1.0e4
$1, \hat{g}_p(1)$	8.1e8 / 1.2e9	1.2e13 / 1.9e13	7.2e10 / 9.7e10	1.0e16 / 1.1e16
$1, \hat{g}_p(1..3)$	7.9e3 / 3.5e4	2.2e6 / 4.4e6	1.1e5 / 3.1e5	3.3e7 / 1.1e8
$1, \hat{g}_p(1..8)$	58 / 1.1e2	3.6e2 / 8.9e2	1.5e2 / 2.8e2	1.7e3 / 3.7e3

Remark 6.3 (what the ledger does and does not reach). It reaches heights up to roughly 10^{13} at $p = 5$ and 10^{15} at $p = 7$ in the two-term case, degrading to a few hundred for eight terms, at $x \leq 8$; no relation was returned below the floor anywhere. It says nothing about relations of larger height, relations involving $\hat{g}_p$ at larger arguments, or — crucially — relations that are not of the form $\hat{G}(\mu) = 0$ for μ of small support. Under Theorem 3.6 the relations that matter for the Kazandzidis algebra are exactly those with $\mu \in \text{im } \delta = \ker m_1$, so the ledger is a search in the right place; it is not a proof of anything.

6.3 What is imported

Paper C's exclusion theorem [3, §5] is the companion negative and we do not repeat its tables: no $\mathbb{Q}$ -affine relation of height $\lesssim 10^4$ between Λ_a (or the Brown–Zudilin analogues) and $\zeta_p(3)$ or $\zeta_p(5)$, at $p = 5, 7, 11, 13$, to 12–27 certified digits; the same for $\log_p 2$, $\log_p 3$, the unit root of $\eta(2z)^4\eta(4z)^4$, and for algebraicity of degree ≤ 3 ; and, across families, no pairwise $\mathbb{Q}$ -linear relation among the Λ_1 's (at $p = 7$, 12–21 certified digits).

Theorem 3.6 does *not* apply to Λ_a , which is an infinite assembly of blocks rather than a finite monomial in tower Betas, and we claim no implication. What the present paper adds to those exclusions is a heuristic, and we label it as one: by Theorems 2.5 and 4.8 the ingredients of Λ_a are exponentials of infinite $\mathbb{Q}$ -combinations of *all* the odd $\zeta_p(m)$ with p -power-weighted coefficients, so a finite low-height relation between Λ_a and a single $\zeta_p(m)$ would be a coincidence of the same species as the ones Hypothesis 3.7 forbids — but it is not one of them, and nothing here proves it cannot happen.

7 The programme

The Λ -algebra is a class of constants with closed forms and no theory. Here are the questions we can now state precisely, in the order we would attack them.

Problem 7.1 (the presentation). Prove Problem 3.10: $\ker \delta$ is generated by the symmetry and trinomial-revision relations. This is a finite-type statement about the free abelian group $\ker m_1$ and is independent of p ; Finding 3.9 confirms it to $N = 11$. With it, Theorem 3.6 becomes a complete determination of the multiplicative relations among the tower Betas, conditional only on (IND_p) . (For the caveat about $\mathcal{E}_p$ itself, see Remark 3.11.)

Problem 7.2 ((IND_p) in weakened form). Hypothesis 3.7 in full is at least as hard as the irrationality of a single $\zeta_p(2k+1)$ for $p \geq 5$. Two weakenings look accessible: (a) prove $\widehat{g}_p(1) \notin \mathbb{Q}$ for some p — equivalently, that $\sum_{m \text{ odd} \geq 3} \zeta_p(m)p^m/(m(1-p^m))$ is irrational; (b) prove that $\widehat{g}_p(1), \dots, \widehat{g}_p(N)$ span a $\mathbb{Q}$ -space of dimension ≥ 2 for some explicit N and p . Both are p -adic Apéry-type questions about a *single* very rapidly convergent series, and (b) is the natural target for a p -adic Ball–Rivoal argument [17, 13].

Problem 7.3 (the Λ -module across bases). Determine the $\mathbb{Q}$ -vector space spanned by $\{\Lambda_a\}_{p \nmid a}$ inside $\mathcal{L}_p$. Paper C excludes low-height relations; Theorem 4.8 shows the a -dependence enters through $h_p(a)$, through $\widetilde{A}_a$ and through the block index set $\{j \leq ap^i\}$. The tractable subquestion: is $\Lambda_a - h_p(a)$ determined by $\Lambda_1 - h_p(1)$ and finitely many Γ_p -values? The block structure says no, and a proof would be the first unconditional non-relation in this subject.

Problem 7.4 (the other fourteen families). Theorems 4.6 and 4.8 are proved for the classical Apéry $\zeta(3)$ pair, using Apéry’s explicit harmonic formula for b_n . Every sporadic B has a harmonic-monomial closed form of weight w [1, §6]; carry the same two estimates through it and one gets $\Lambda_a \in \mathcal{L}_p$ for all fifteen families, with $\mathcal{E}_p$ enlarged by the character values. What is the correct twisted analogue of $h_p^{(w)}$, and does χ enter the blocks or only the normalisation?

Problem 7.5 (a weight filtration). Is there a filtration on $\mathcal{L}_p$ under which $h_p^{(w)}(a)$ has weight w , $\beta_p(B, D)$ has weight 0, and Λ_a has weight w ? The obstruction is that β_p is an exponential of a series mixing all odd weights; a filtration would have to be by the *order of vanishing* of $\widehat{g}_p$ at $x = 0$, which is 3 — the same for every tower Beta, so the filtration would be trivial. A graded object, if one exists, is the right home for a motivic statement.

Problem 7.6 (what a period statement would mean). The archimedean Apéry limits are periods. Their p -adic counterparts are values of $\prod_{k \geq 1} G(xp^k)$ with $G = \Gamma_p e^{-\Gamma'_p(0)x}$, the same generating function whose Taylor coefficients are the Beukers–Vlasenko Frobenius structure constants [8]; archimedeanly the analogous generating function is the one of Bloch–Vlasenko [7].

Is there a category in which $\prod_{k \geq 1} G(xp^k)$ is the period of an object, and in which the divisor map δ of Definition 3.3 is the shadow of a coproduct? The relations (3.3), (3.4) are exactly those of a $\mathbb{Z}$ -valued “divisor of a binomial”, which is suggestive.

Problem 7.7 (falsifiable next computations). Three, in increasing cost.

- (1) Extend Finding 3.9 to $N = 20$; a failure there would refute Problem 7.1 outright. Cost: exact Smith normal forms of matrices of size $\binom{20}{2} = 190$.
- (2) Extend Finding 6.2 to $x \leq 16$ and $A = 60$ certified digits. $\widehat{g}_p$ costs only $\zeta_p(m)$ for $m \leq A+6$, so this is cheap; the discriminating quantity is whether any height fails to scale with A .
- (3) Compute Λ_a for the twisted families to 18 digits and test $\Lambda_1^{(\text{fam}_1)}/\Lambda_1^{(\text{fam}_2)}$ against $\mathcal{E}_p$ -monomials rather than against $\mathbb{Q}$ — Paper C tested the latter only, and Theorem 3.6 says $\mathcal{E}_p$ is the right group to quotient by.

A Reproduction

All computations use exact integer or rational arithmetic; no floating point occurs at any stage. The programs accompanying this paper are:

- `lam.py` — the p -adic number class with an honest precision bound carried through every operation; $\widehat{g}_p$, β_p and E_p from the Kubota–Leopoldt values $\zeta_p(m)$; the direct Kazandzidis limits $c_p(A, B)$ from exact binomial residues by Legendre/Morita splitting (one linear pass per prime, recording only the needed factorial-unit indices); the verification of (2.6) (Finding 2.7); the three relation sweeps (Finding 3.2); the (IND_p) hunt with the planted-relation validation (Findings 6.1, 6.2); and the exact integer linear algebra for Finding 3.9, including the assertion that every proposed generator really lies in $\ker \delta$; and the tower-harmonic constants $h_p^{(w)}(a)$, with the distribution relation and the relation hunt of Finding 4.4 (the index i ranging over all integers with $ap^i \geq 1$, cf. Remark 4.3).
- `atil.py` — the block-by-block check of Theorem 4.6 for $a = 1, 2$ (Finding 4.11), with a_{ap^s} built independently as an exact integer from the defining double sum.

The $\zeta_p(m)$ implementation, its validation record and the exact LLL are those of [3, App. A] and are re-used unchanged; the certified precision of every reported digit is carried by the number class rather than asserted.

Acknowledgements

[TO BE SUPPLIED]

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